1) Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

- Task: Simulated an incorrect hostname mapping using the Windows hosts file.
- Incorrect IP: 192.0.2.1
- Command used: ping www.spotify.com
- Expected observation: The hostname resolves to 192.0.2.1, and the ping may time out.
- Conclusion: An incorrect hosts-file entry can override the normal DNS resolution path and cause connectivity problems even when public DNS is working.

2) Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

1. Create the network topology

Use these devices in Cisco Packet Tracer:

DHCP Server

192.168.1.2

Switch

Client PC

DHCP enable

Connect the DHCP server and PC to the switch using copper straight-through cables. Add a router if you want to test connectivity to another network.

Use this addressing plan:

Setting Value

Network 192.168.1.0/24

DHCP server 192.168.1.2

Correct gateway 192.168.1.1

Incorrect gateway for the test 192.168.1.254

DHCP client range 192.168.1.100 onward

2. Configure the DHCP server

1. Click the Server device.

2. Select Desktop → IP Configuration.

3. Set its static IP configuration:

- IP address: 192.168.1.2

- Subnet mask: 255.255.255.0

- Default gateway: 192.168.1.1

4. Open Services → DHCP and turn the DHCP service On.

5. Create a DHCP pool with these settings:

DHCP setting	Value
Pool name	OFFICE-LAN
Default gateway	192.168.1.254
DNS server	8.8.8.8
Start IP address	192.168.1.100
Subnet mask	255.255.255.0
Maximum users	50

6. Click Add or Save, depending on your Packet Tracer version.

The default gateway 192.168.1.254 is intentionally incorrect for this test. The correct router address should be 192.168.1.1.

3. Connect the client PC

1. Click the PC.

2. Select Desktop → IP Configuration.

3. Select DHCP.

4. Wait for the DHCP assignment.

3) Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Symptom	Likely cause	Solution
IP access works, domain lookup fails	DNS configuration problem	Verify DNS server address
DNS request times out	Server unreachable or not responding	Check server status, routing, and firewall
NXDOMAIN	DNS record does not exist	Create or correct the record
Direct query works, default query fails	Incorrect client DNS settings	Set the correct DNS server
DNS resolves but website does not open	Possible web service, routing, or firewall issue	Test connectivity and the destination service

4) Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

. Common DNS issues

DNS issue	Symptoms	Possible causes	Recommended solutions
1. DNS server not responding	Websites do not open by name; nslookup times out.	DNS service is stopped, server is unreachable, or firewall blocks port 53.	Check DNS service status, network connectivity, and firewall rules. Restart the DNS service if needed.
2. Incorrect DNS record	A domain resolves to the wrong IP address.	Incorrect A record or outdated DNS data.	Correct the DNS record, increment the zone's SOA serial if applicable, and reload the zone.
3. Missing DNS record	nslookup returns NXDOMAIN for a hostname.	The hostname or record does not exist in the DNS zone.	Add the required record and validate the zone configuration.
4. Incorrect client DNS settings	Some domain names fail to resolve on a client.	The client is configured to use an incorrect DNS server.	Check ipconfig /all and configure the correct DNS server address.

2. Common DHCP issues

DHCP issue	Symptoms	Possible causes	Recommended solutions
1. Client cannot obtain an IP address	Client receives an APIPA address such as 169.254.x.x on Windows.	DHCP server is down, unreachable, or DHCP requests are blocked.	Check DHCP service, network connections, VLAN configuration, and DHCP relay if needed.
2. DHCP scope exhausted	New clients cannot obtain an IP address.	All available addresses in the DHCP pool have been leased.	Expand the address pool if safe, remove stale leases, or reduce lease duration where appropriate.
3. Incorrect default gateway	Client can communicate locally but cannot reach other networks.	DHCP pool advertises the wrong gateway address.	Correct the DHCP gateway option and renew the client's lease.
4. Incorrect subnet mask	Some local devices are unreachable or traffic follows an incorrect route.	DHCP supplies an incorrect subnet mask.	Correct the subnet mask in the DHCP scope and renew the lease.